Introduction to Circular Convolution and Filtering with the DFT

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Key Concepts

1) The DFT can be used to filter data by multiplication in the frequency domain provided the data are processed in blocks and the convolution-multiplication property is properly applied.

2) Multiplication of length-N DFT coefficients corresponds to convolution of one time signal with an N-periodic version of the other.

    a) This is called circular convolution.

    b) Recovery of the ordinary or linear convolution from circular convolution requires that the DFT length, N, exceed the length of the linear convolution of the signals.

3) Linear convolution of length-M_h and length-M_x signals is length $M_h + M_x - 1$, so we require $N \geq M_h + M_x - 1$ to recover the linear convolution from the circular convolution computed with the DFT.

Video Screenshots

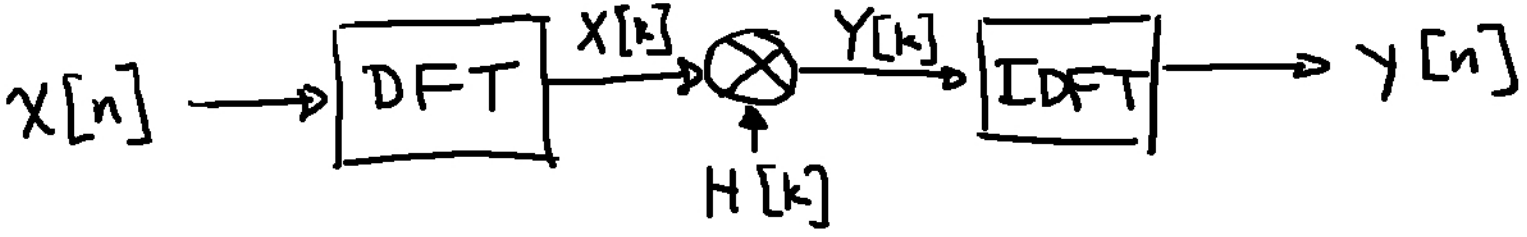


$$y[n] = h[n] * x[n]$$

$$\updownarrow \text{ DTFT}$$

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$$

Can we filter with DFT? $Y[k] = H[k] \cdot X[k]$



Issues

- DFT requires a block of data
- DFT convolution-multiplication property

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Block processing -

must have $\{x[0], x[1], \ldots x[N-1]\}$ to compute $X[k]$

introduces time lag in real-time applications

Multiplication of DFT coefficients

$$x[n] \xleftrightarrow{N\text{-DFT}} X[k] \quad , \quad h[n] \xleftrightarrow{N\text{-DFT}} H[k]$$

$$Y[k] = H[k]X[k] \xleftrightarrow{N\text{-DFT}} \tilde{y}[n] = \underbrace{h[n] * \sum_{\ell=-\infty}^{\infty} x[n-\ell N]}_{\text{"circular convolution"}}$$

(N-periodic)

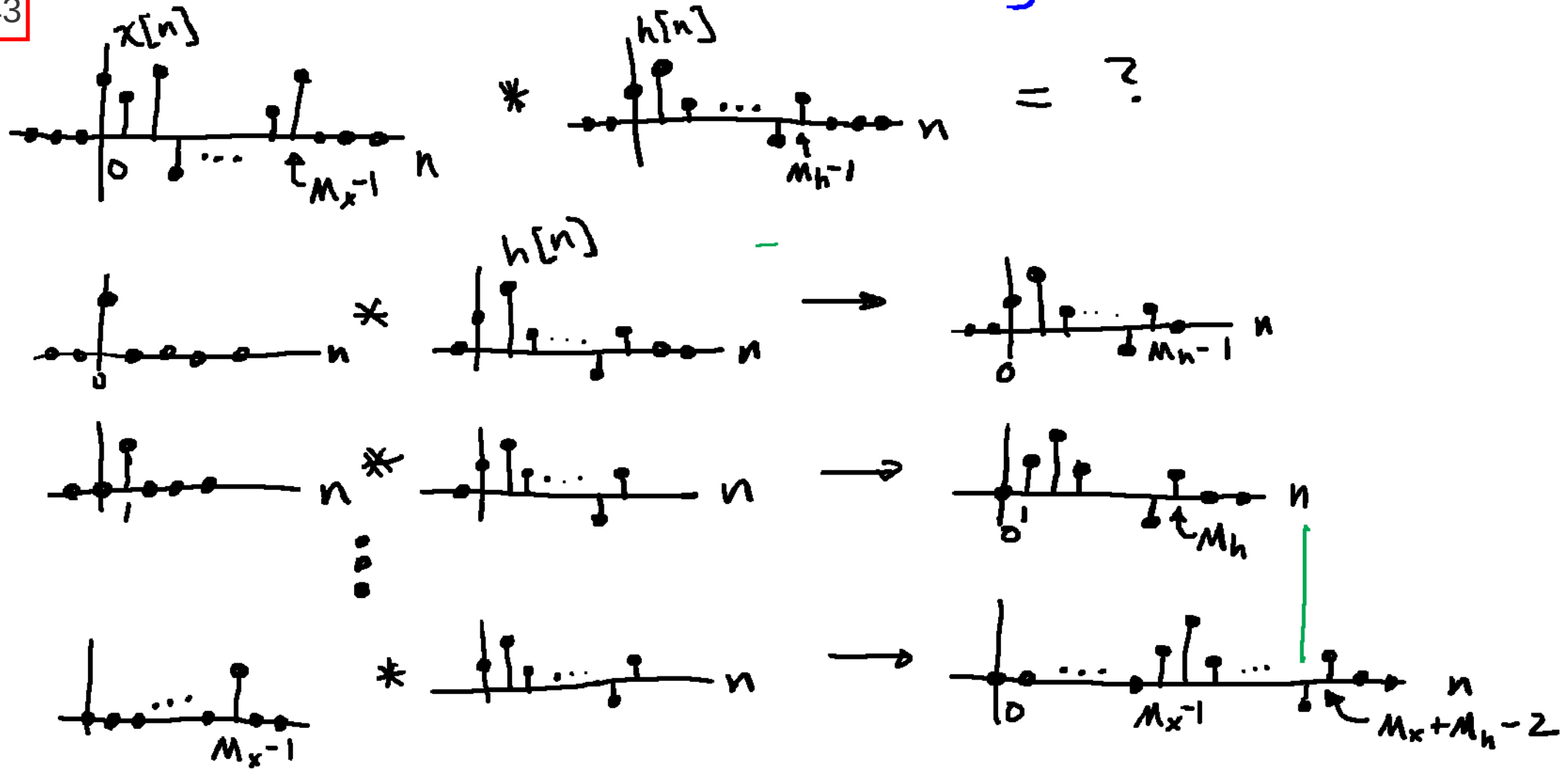
To recover $y[n] = h[n] * x[n]$

$h[n]*x[n]$ must not overlap with $h[n]*x[n+N]$, $h[n]*x[n-N]$,

$h[n]*x[n+2N]$, $h[n]*x[n-2N]$, ...

Duration of convolution is key

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$x[n] * h[n]$ has duration $M_x + M_h - 1$

$h[n] * x[n]$: $M_h + M_x - 1$ duration

$h[n] * x[n - lN]$ is $h[n] * x[n]$ shifted by lN

Obtain $y[n] = h[n] * x[n]$ from $Y[k] = H[k]X[k]$

- require $N \geq M_x + M_h - 1$ (circular conv = linear conv)

pad $h[n]$ and $x[n]$ with zeros to length N

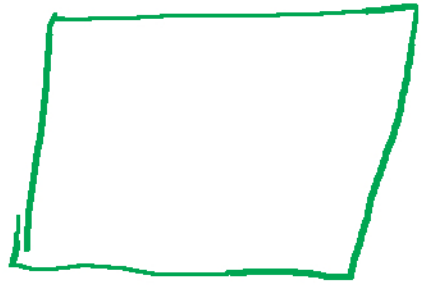
Fast convolution –

FIR $h[n]$

Segment $x[n]$

Take advantage of FFT algorithm

Construct $y[n]$ from segment outputs



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